An actual BK torsion-field criterion for every degree

For the actual E/F_5: $y^2=x^3+1$ and r=3, Lean proves that full geometric 3-torsion descends to the actual point image of a field of size 5^k exactly when 3 divides 5^k-1, equivalently when k is even. For each positive k, it constructs an actual embedding of Mathlib's GaloisField 5 k and proves the criterion for that explicit field. The actual curve has six points; all source arithmetic hypotheses, including r not dividing q−1, are proved. No torsion basis, determinant or Weil-pairing theorem is assumed. [pcc:unverified]

The actual point P=(1,beta), $\text{beta}^2=2$, has exact order three and Frobenius(P)=−P≠P. It supplies an actual nonrational witness for every odd degree. Every actual 3-torsion point is fixed by Frobenius squared. [pcc:unverified]

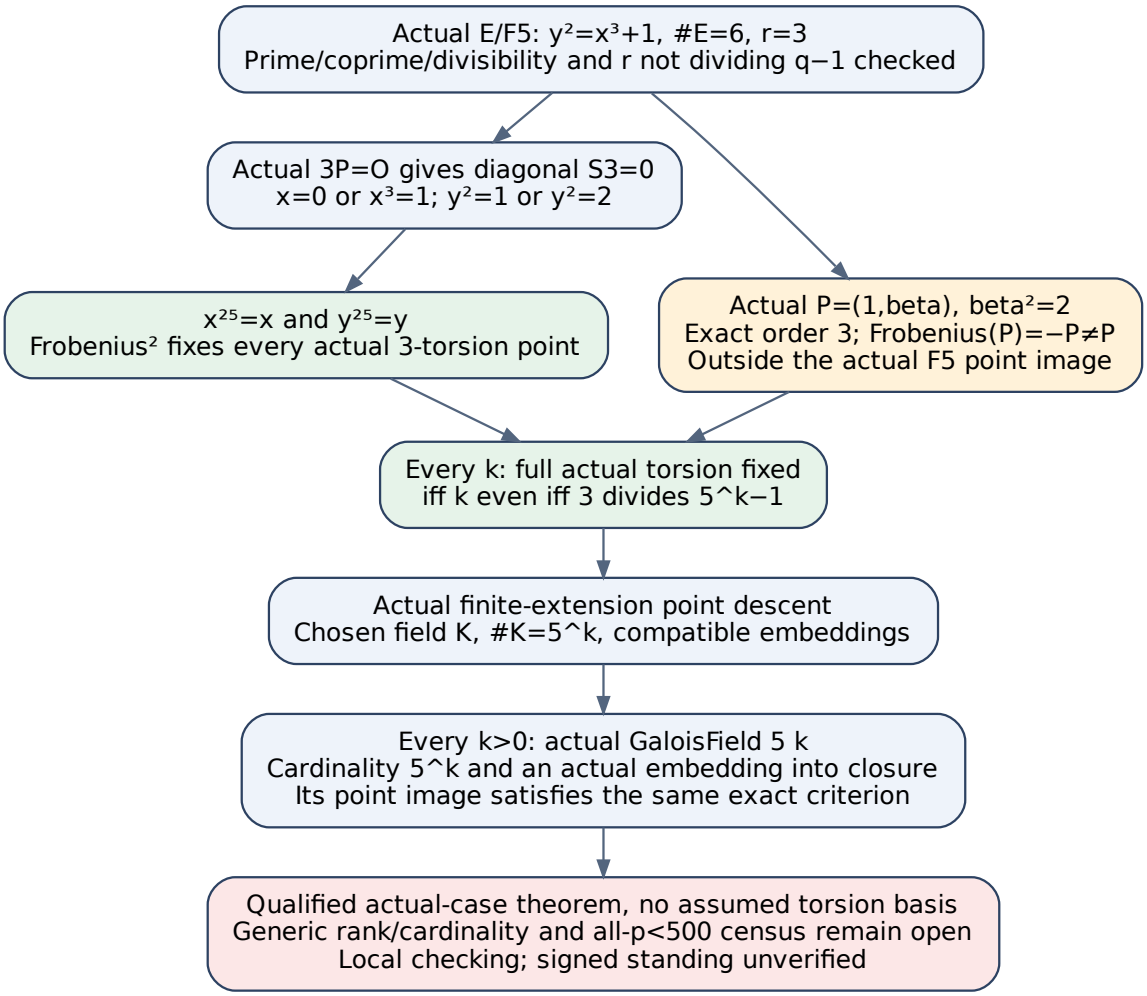


Figure 1: Actual coordinates, all-degree criterion and explicit finite-field construction. [pcc:unverified]

Proof obligations, scope and skeptical controls

If 3P=O for an actual nonzero point (x,y), then 2P=−P. The checked actual repeated-S3 bridge gives diagonal S3 zero. Its expression $-3x^4-12x$ factors in characteristic five to x=0 or $x^3=1$. The actual curve equation gives $y^2=1$ or $y^2=2$. Therefore $x^{25}=x$ and $y^{25}=y$: actual Frobenius squared fixes every geometric 3-torsion point, including infinity. [pcc:unverified]

An actual algebraic-closure point (1,beta), $\text{beta}^2=2$, has order three by the same repeated-S3 group bridge. Its fifth power relation $\text{beta}^5=-\text{beta}$ proves it is moved by Frobenius and is outside the actual F_5 point image. Even iterates fix all torsion; every odd iterate moves this witness. The order of 5 modulo 3 is two, giving the arithmetic equivalence. [pcc:unverified]

The existing actual finite-extension theorem identifies Frobenius fixedness with membership in an actual point-map image. The new construction uses GaloisField 5 k, its proved cardinality 5^k and IsAlgClosed.lift to an actual algebra embedding, with scalar-tower compatibility checked. It requires k>0. The separate iterate theorem includes k=0 as an ordinary identity; no field with one element is asserted. [pcc:unverified]

Module / declarations	Exact checked obligation
`ThreeTorsion`: `diagonal_of_three`, `three_coordinate_classification`	Actual torsion equation implies diagonal S3 and the x classification
Three: `actual_equation`, `frobenius_square_fixed`	Actual y equation, coordinate powers and Frobenius-square identity
`QuadraticWitness`: `P_order`, `P_frobenius_neg`, `P_not_rational`	Actual exact-order-three point, anti-Frobenius action and nonrational image
`EmbeddingCriterion`: `all_torsion_fixed_iff_even`, `divisibility_iff_even`	All-k torsion criterion and exact arithmetic equivalence
Embedding: `actual_extension_torsion_criterion`	Actual finite-extension point image, with its stated compatible embeddings
`GaloisFieldCase`: `actual_galois_field_criterion`	Explicit actual finite field and algebra embedding for every positive k
`BKCertificate`: `bk_actual_case_audit_certificate`	All source hypotheses, witness, all-k and explicit-field conclusions

The final whole gate checks 37 declarations, zero failing and zero trusted-native exceptions. The earlier 36-declaration gate is retained. Lean 4.35.0 / Mathlib 56a00585; only propext, Classical.choice and Quot.sound. No holes, custom axioms, native_decide, linter waivers or added elaboration settings. All 29 attempt files and both gate receipts are archived losslessly. Source commit `11189a29` precedes packet `PCC-20261110-99983d`. Actual local verdict and signed standing are retained in verification.json; signed standing remains unverified. [pcc:unverified]

Original BK Theorem 1 conditions/proof/remarks, pages 141–143, were reread; page 142 was visually inspected. Primary PDF SHA-256:

`ac837b8edc47b346dbb341f787c41de0402c95371a159f9786b747bb2c987499`

The source's r^2-point-count statement and the user's geometric-torsion inclusion form are distinct formal obligations. This study proves the latter for one actual curve at every degree. Generic rank/cardinality, determinant, Weil necessity, actual torsion cardinality nine and the requested all-p<500 census remain unclaimed. [pcc:unverified]

The prior actual F_7/#9/nonrational 3-torsion witness remains the negative control for removing r not dividing q−1. It does not refute the qualified source theorem. The prior scalar S3 supplement IDC1h392e1740fcb7cf05 is copied unchanged; its sampled polynomial identity does not prove these new curve/descent results. No fresh sampling, native experiment, timing or paid compute. KB/Conductor unavailable; primary/repository/Mathlib fallback. No novelty claim. [pcc:unverified]

Original BK paper. Source pins, actual verdict, proof map and archives: study sources.json, verification.json and README.md. [pcc:unverified]